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Assignment Word Embeddings – Theory and Analysis

1. What is an Word Embedding?

A word embedding is a way of representing a word as a dense, low dimensional, real valued vector (typically 50-300 dimensions) such that words with similar meaning or usages end up close together in the vector space.
Embeddings are learned from large text corpora, usually based on the distributional hypothesis: "a word is ~~ea~~ characterized by the company it keeps" (First 1957). ie words that occurs in similar contexts tend to have similar meanings.

2. One-Hot Encoding vs Dense Word Embeddings

Aspect	one Hot Encoding	Dense Word Embeddings
Vector size	Equal to vocabulary size (v) eg 50000	Small fixed (eg:- 50-300)
Sparsity	Sparse - exactly one "1", rest "0"	Dense - every value is (usually) non-zero
Semantic meaning	None all words are equidistant	Captures meaning - Similar words are close.

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Aspect	one Hot Encoding	Dense Word Embedding
~~vectors~~:		
Similarity	Cosine similarity between any two one hot vectors is always 0	Cosine similarity reflects real semantic closeness.
Storage/ Compute	Expensive for large vocabularies	Efficient, Compact
Example	"cat" = [0,0,1,0...0], "dog" = [0,1,0,0...0] Orthogonal	"cat" = [0.21, -0.4, 0.85 ...], "dog" = [0.19, -0.38, 0.81 ...] close vectors

Note: one hot encoding treats words as atomic symbols with no relationship to each other.

Dense embeddings place words in a continuous space where geometic distance / direction encodes semantic and syntactic relationships.

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3. Word2Vec : CBOW vs Skip Gram

Word2Vec (Mikolov et al., 2013) learns embeddings via a shallow network (neural) trained on a prediction task.

It has two architectures:

3.1 CBOW (Continuous Bag of Words)

- Task: Predict the center (target) word given its surrounding context words.
- Input: average/sum of context word vectors → Output: probability distribution over vocabulary for the center word.
- Faster to train, works well for frequent words, smooths over a lot of distributional information (because it averages context.)

Examples: · Context = ["the", "cat", "on", "the", "mat"], missing word = "sat" → Input: (the, cat, on, the, mat) → Output: "sat"

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~~3.2 Skip Gram~~

- ~~Task: predict the next Context words (reverse of CBOW)~~

- ~~Input single~~

3.2 Skip Gram

Task: Predict the context words given the center word (reverse of CBOW)

Input: ~~Sigr~~ Single center word → Output: probability distribution prediction each surrounding word.

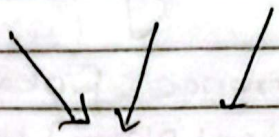
Slower to train but works better for rear words and small datasets, because each word gets more individual training signal

Example: Center word = "Sat" → Predict, "The", "Cat", "On", "the", "mat"

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CBOW

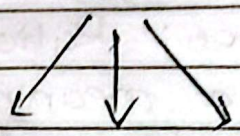
the on mat



Cat?

Skip Gram

Cat



the? on? Mat?

Negative Sampling

Both architectures use negative sampling to make training efficient: instead of computing a full softmax over the entire vocabulary (very expensive), the model learns to distinguish the true context word from a small number of randomly sampled "negative" (incorrect) words. This turns a multi class problem into several binary classification problems.

4 Glove (Global vectors)

- Developed by Pennington, Socher, and Manning (Stanford 2014)
- Unlike word2vec (which is predictive, learning from local context windows), Glove is count based: it first builds a global word - word co occurrence matrix over the entire corpus,

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then factorizes it so that the dot product of two word vectors approximates the log of their co-occurrence probability.

- Key idea: Ratios of co-occurrence probabilities encode meaning. Eg P(ice | steam) vs P(ice | solid) helps distinguish ice from unrelated words.

- Combines the advantages of global matrix factorization (like LSA) with local context window methods (like Word2Vec)

5. Fast Text

- Developed by Facebook AI Research (Bojanowski et al. 2017)

- Extends Word2Vec by Representing each word as a bag of character n-grams (subword units), plus the whole word itself.

- Example: "apple" (n=3) → ⟨ap, app, ppl, ple, le⟩, ⟨apple⟩

- The word's embedding = sum of its ngram embeddings

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- Advantage: Can generate embeddings for out of vocabulary (OOV) words and handles morphologically rich language (eg: prefixes/ suffixes) much better than Word2Vec or Glove, Since it can compose an embedding for an unseen word from known subword pieces.

6. Comparison Table

Feature	Word2Vec	Glove	Past Text
Approach	Predictive (local context window)	Count based (global Co-occurrence)	Predictive, Subword (Character n-grams)
OOV word handling	No	No	Yes
Captures morphology	No	No	Yes
Training speed	Fast	Moderate (needs co occurrence matrix)	Slower (more subword computations)
Good For	General semantics task	Capturing global Corpus Statistics	Rare words, misspellings, morphologically rich language

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7 Semantic similarity in Embedding Space

Measured by Cosine similarity!

$$\cos\theta = (A \cdot B) / (\|A\| \, \|B\|)$$

→ Range: -1 (opposite) ..... +1 (same direction)

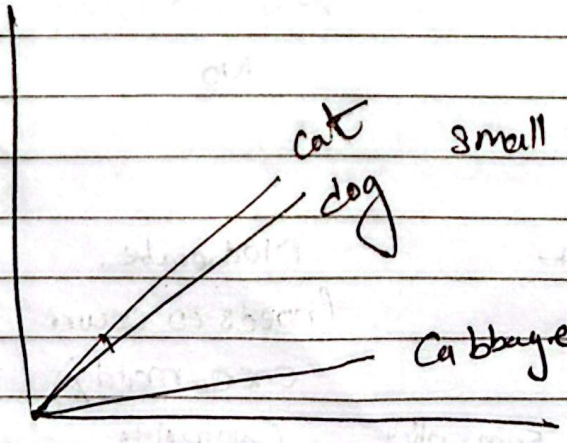
→ words used in similar contexts → high cosine similarity.

Worked example (real 50-d Glove vectors)

Cos(King, queen) ≈ 0.78 → high (both royal concepts)

Cos(Cat, dog) ≈ 0.92 → very high (both common pets)

Cos(King, cabbage) ≈ 0.11 → low (unrelated)



small θ → cosθ near 1

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direction matters not length - that is why cosine is used instead of plain distance

fig 4 - angle between word vectors = semantic closeness

vector arithmetic (analogies)

Embeddings capture ~~relate~~ relations as consistent directions in space:

King - man + women ≈ queen

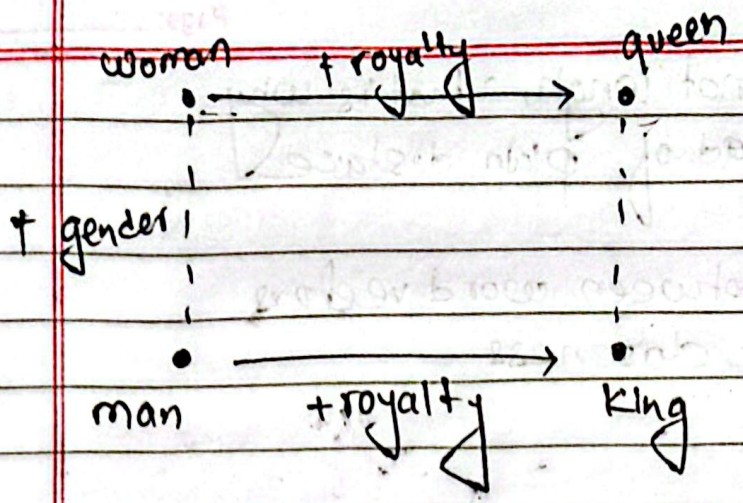
→ The offset (King - man) encodes a royalty direction adjusted for gender".

→ Adding "women" moves us to the female royal region → lands near queen.

- on real Glove vectors the top result was indeed queen ($\cos \approx 0.85$)

- Followed by throne, prince, daughter — all close royal terms.

- Another: paris - france + Italy ≈ rome (Europ. capital - country direction)



Two offsets are parallel → relations = directions

Fig 5- the analogy parallelogram

8. Role of Dimensionality

→ low dims (25-50) compact, fast - but miss fine grained distinctions. Several senses get compressed together, nuance is lost

→ Higher dims (200-800): Capture subtler syntactics + semantics distinctions, but:

- need more training data (else overfitting/noise)
- Cost more money & computation

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- give · ~~diminst~~ diminishing ~~rate~~ resty after a point.

- 100 - 500 · dims = ' common sweet spot for many NLP tasks.

→ Tade off ! dimensionality is a hyper perameter balancing ~~expen~~ expressiveness against ~~effect~~ efficiency + generalisation.

- very high d also shows ~~neerst~~ nearest neighbour search and dilutes meaningful distances among near - random ones. (curse of ~~dimonth~~, dimensionality)

- PCA and t-~~SNE~~ SNE project high d embedding ~~two~~ down to 2D/3D So humans can inspect the clusters

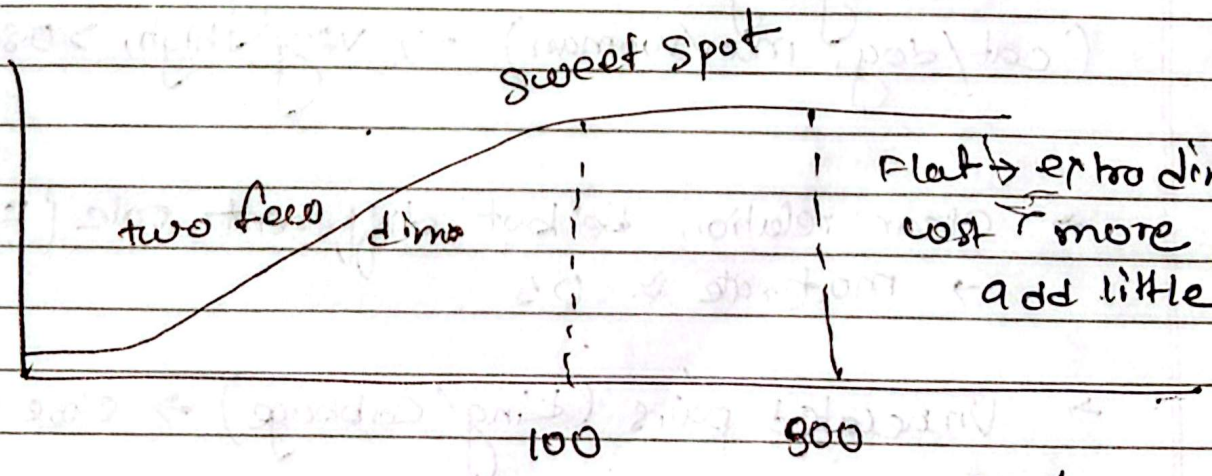


Fig 6: expressiveness vs cost

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Embedding Space Exploration

1. Cosine Similarity Computations

$$\cos\theta = (A \cdot B) / (\|A\| \|B\|)$$

word pair	Cos. sim.	Interpretation
King, Queen	0.7839	Strongly related - both royalty terms
man, women	0.8860	very strong - paired gender terms, co-occur often
Cat, dog	0.9218	extremely high - near-identical contex

Observation

→ Near Synonyms or words sharing contexts (cat/dog, man/woman) → very high, > 0.85

→ Clear relation but different role (King/man) → moderate ≈ 0.5

→ Unrelated pairs (King/Cabbage) → close to 0.

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→ Note: happy / joyful is only 0.56 - true synonyms can score lower than merely co-occurring words, because embeddings measure context, not dictionary meaning.

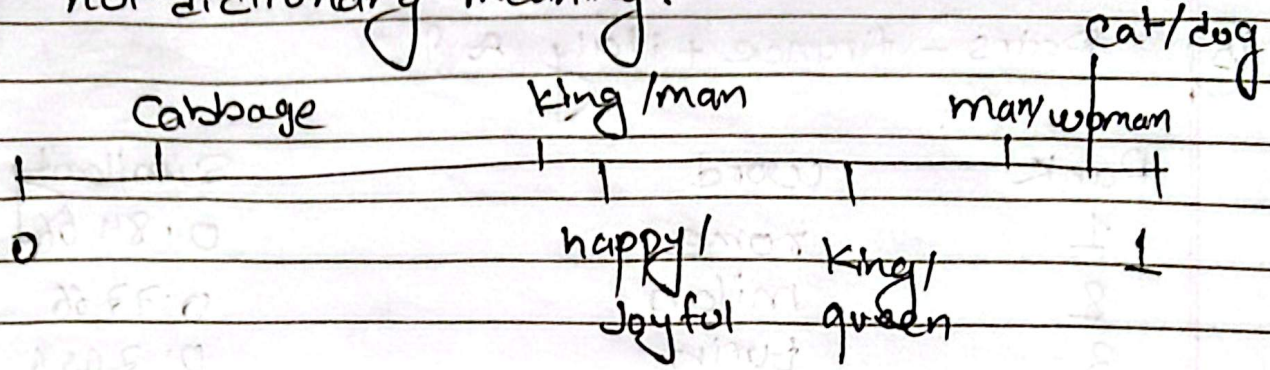


fig: all seven pairs placed on the ~~similarity~~ Similarity Scale

2. Vector Arithmetic — Analogy Task

(a) King - man + woman ≈ ?

Rank	word	Similarity
1	queen	0.8524
2	throne	0.7664
3	prince	0.7592
4	daughter	0.7474
5	elizabeth	0.7460

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Classic analogy holds - remove the "male" direction, add the "female" one, and we land on queen.

(b) 'Pairs - france + italy ≈ ?

Rank	word	Similarity
1	rome	0.8468
2	milan	0.7766
3	turin	0.7666
4	venice	0.7582
5.	madrid	0.7566

The "capital of" relation transfers: France → Paris become Italy → Rome.

Ranks 2-4 are all Italian cities → the arithmetic also carries the country information.

(c) Walked - walk + swim ≈ ?

Rank	word	Similarity
1	swam	0.7987
2	rowed	0.6911

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3 paced 0.6769

Even a purely grammatical relation (present → past tense) is stored as a consistent vector offset - not only meaning relations.

(B) Visualization - PCA and t-SNE

→ 25 words from 5 semantic categories: royalty, animals, geography / capitals, fruits, emotions.

→ ~~Re~~ Projected from 50-d Glove space down to 2-d by two methods

- PCA - linear, maximises variance along orthogonal axes.

- t-SNE - non-linear, preserves local neighbourhood ~~Struct~~ structure.

- PCA explained variance (first 2 components)

≈ 25% + 17% ≈ 42%

- So a 2-d picture must lose information

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- yet the main groupings still seperate cleanly

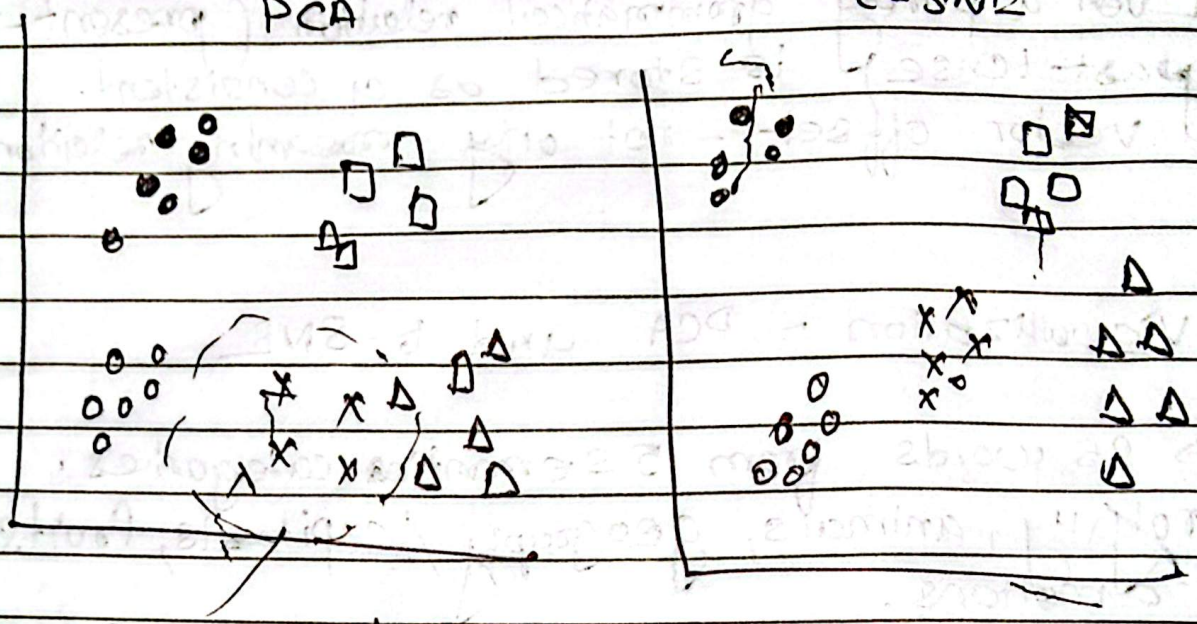


fig: 25 words, 5 categories, projected to 2d

4. Interpretation of the clusters

Royalty - king, queen, prince, princess, throne

→ Tightly grouped in both plots - heavy contextual overlap (royal court, monarchy, narratives)

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Animals - cat, dog, lion, tiger, elephant

→ Own distinct region, well away from royalty & fruit - a different semantic domain (pets and wildlife)

Geography - France, paris, italy, rome, spain, madrid

- Clearly sp separated cluster.
- Interesting: each country sits next to its own capital - the capital-of relation shows up geometrically, not only through arithmetic

Fruits - apple, banana, orange, grape, mango

- Tight, isolated group - fruit names share contexts (recipes, grocery lists) and rarely meet royalty or animals.

Emotions - happy, sad, angry, joyful, furious

- Clusters too, but more spread out than the rest.

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- Reason: emotion words appear in every topic - any subject can be described as happy or sad
- so they carry more contextual variety, hence more spread.

Aspect	PCA	t-SNE
type	Linear projection	non-~~linear~~ linear embedding
Optimises	global variance on orthogonal axes.	local neighbourhood structure
Result here	group visible but some overlap	tighter, more separated clusters
Example	"orange" drifts near emotion group	"orange" stays with the fruits
Caution	distances are meaning fol	distances between clusters mean little

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→ t-SNE looks cleaner because it explicitly optimises to keep neighbours together; PCA only stretches the data along its highest-variance directions.

→ Hence PCA can place conceptually distinct groups near one another.

Conclusion: Even a modest 60 dimensional Glove space organises words into semantically coherent neighbourhoods with no explicit supervision - purely from co-occurrence counts in a large corpus. Strong evidence that embedding space encode meaning as geometric structure.